

NIKOLAY TULUPOV

Radio Systems & Electronics Engineer

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PROFILE

Radio systems and electronics engineer with over two years of experience in end-to-end development of hardware–software systems. Designed a remote-control system for a Motorola XTL2500 radio, covering circuit and PCB design, embedded interfaces, the SB9600 protocol, real-time bidirectional audio, Linux software, system integration, testing, and technical documentation. Experienced with radio equipment, measurement instruments, single-board computers, and microcontrollers.

CORE EXPERTISE

Radio & Electronics: radio equipment, circuit design, schematic capture, PCB design, prototyping, hardware debugging, test instruments

Embedded & Interfaces: microcontrollers, single-board computers, UART, SPI, I2C, RS-232, GPIO, ADC/DAC, CAN

Software & Systems: C/C++, Python, Linux, CMake, TCP/IP, client–server systems, real-time audio, Git

Engineering Tools: EasyEDA, MATLAB/Simulink, Motor-CAD, Xilinx ISE, oscilloscope, soldering equipment

RELEVANT EXPERIENCE

TIENDA LLC

Apr 2026 – Present

Engineer, Technical Department

Moscow, Russia

- Continue development of the Motorola XTL2500 remote-control system following the project’s organizational transfer from PromSvyazRadio.
- Develop and debug its hardware and software components and maintain project technical documentation.

PromSvyazRadio LLC

Feb 2024 – Apr 2026

Assistant to the Chief Engineer, Technical Department

Moscow, Russia

- Developed a distributed client–server system on Linux-based single-board computers for remote control of a Motorola XTL2500 radio.
- Implemented real-time bidirectional audio streaming, command handling, and interaction with the radio through the SB9600 protocol.
- Parsed SB9600 frames and radio-display data and integrated UART, SPI, I2C, ADC/DAC, and network interfaces.
- Designed the system schematic and PCB and contributed to the enclosure, assembly, testing, and technical documentation.
- Performed end-to-end hardware/software integration and debugged the system using laboratory measurement equipment.

AmpereMagneTe

Mar 2025 – Mar 2026

Engineering Intern, Artificial Intelligence Group

Troitsk, Moscow, Russia

- Developed embedded firmware for an AT32-based power inverter and software for remote inverter control and test-bench operation.
- Implemented CAN-based telemetry acquisition, visualization, logging, and experimental data analysis.
- Co-developed StandControl, a registered client–server application for remote control and monitoring of a power inverter.

ADDITIONAL ENGINEERING EXPERIENCE

Moscow Institute of Physics and Technology

Mar 2026 – Present

Research Engineer, Robotics and Electric Drive Laboratory

Moscow, Russia

- Develop software for experimental test-bench operation, data processing, visualization, and automated engineering calculations.
- Model electric-drive systems and experimentally validate control algorithms on physical equipment.
- Automate Motor-CAD calculations and electric-motor geometry optimization.

SELECTED ENGINEERING PROJECTS

Remote-Control System for the Motorola XTL2500 Radio

- Designed an end-to-end hardware–software system for operating a radio remotely over an IP network.
- Developed C/C++ software for Linux-based single-board computers, including client–server communication and real-time bidirectional audio using ALSA.
- Implemented SB9600 command exchange, frame parsing, display decoding, and integration with UART, SPI, I2C, ADC/DAC, and Ethernet interfaces.
- Designed the schematic and PCB and participated in enclosure design, system assembly, hardware debugging, and documentation.

StandControl: Power-Inverter Control and Monitoring System

- Co-developed a client–server application for remote inverter control over CAN, including torque, speed, and current operating modes.
- Implemented telemetry monitoring, visualization, logging, network communication, and error handling using Python and C/C++.
- Registered as a computer program in 2026 (No. 2026614166).

OptiRunMotorCAD

Developed a software application for automating Motor-CAD calculations and electric-motor geometry optimization. Registered as a computer program in 2025 (No. 2025693472).

EDUCATION

Moscow Institute of Physics and Technology

2026 – 2028 (expected)

M.Sc. in Applied Mathematics and Physics

Moscow, Russia

Specialization: E-Mobility – Technologies, Materials and Control

Moscow Institute of Physics and Technology

2022 – 2026

B.Sc. in Applied Mathematics and Physics

Moscow, Russia

Concentration: Radio Engineering and Computer Technology

SELECTED PUBLICATIONS & RECOGNITION

- N. Tulupov and D. Kanurin, “Physics-Informed TD3 for End-to-End Torque Control of a PMSM Drive,” 2026 IEEE 27th International Conference of Young Professionals in Electron Devices and Materials (EDM), 2026.
- Best Presentation Award, Exact and Engineering Sciences section, III International Youth Conference “AI Technologies in Science and Education,” Moscow State University, 2025.

LANGUAGES

Russian – Native | English – B2 (Upper-Intermediate)